SUM TO N TERMS OF SPECIAL SERIES

JEE Advanced Preparation — Mathematics

Standard Results | Product Series | Factorial Sums | Vn Method | Second-Difference Sequences

Subject	Mathematics
Level	JEE Advanced
Topic	Sum to N Terms — Special Series
Prerequisite	Basic algebra, AP/GP, Factorial notation

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1. STANDARD RESULTS — COMPLETE DERIVATIONS

1.1 Sum of First n Natural Numbers (S_1)

Let S_1 = 1 + 2 + 3 + ... + (n-1) + n ...(i)

Reversing: S_1 = n + (n-1) + ... + 2 + 1 ...(ii)

Adding (i) and (ii) term-by-term:

$2S_1$ = (1+n) + (2+n-1) + ... = n(n+1)

Result: S_1 = n(n+1)/2

Examples: $S_1(10) = 55$, $S_1(100) = 5050$, $S_1(n) = n(n+1)/2$

1.2 Sum of Squares of First n Natural Numbers (S_2)

Method of Differences: Start with the identity:

r^3 - (r-1)^3 = 3r^2 - 3r + 1 [expand (r-1)^3 and simplify]

Write this for r = 1, 2, 3, ..., n and sum both sides:

r=1: 1^3 - 0^3 = 3(1)^2 - 3(1) + 1

r=2: 2^3 - 1^3 = 3(2)^2 - 3(2) + 1

... ...

r=n: n^3 - (n-1)^3 = 3n^2 - 3n + 1

LHS telescopes: n^3 - 0 = n^3

RHS: 3*Sum(r^2) - 3*Sum(r) + n = $3S_2$ - 3*n(n+1)/2 + n

So: n^3 = $3S_2$ - 3n(n+1)/2 + n

=> $3S_2$ = n^3 + 3n(n+1)/2 - n = [2n^3 + 3n(n+1) - 2n] / 2

=> $3S_2$ = [2n^3 + 3n^2 + 3n - 2n] / 2 = [2n^3 + 3n^2 + n] / 2 = n(2n^2+3n+1)/2

=> $3S_2$ = n(2n+1)(n+1)/2

Result: S_2 = n(n+1)(2n+1)/6

Examples: $S_2(5)$ = 5*6*11/6 = 55, $S_2(10)$ = 10*11*21/6 = 385

1.3 Sum of Cubes of First n Natural Numbers (S_3)

Method of Differences: Use the identity:

r^4 - (r-1)^4 = 4r^3 - 6r^2 + 4r - 1

Sum from r=1 to n (LHS telescopes to n^4):

n^4 = $4S_3 - 6S_2 + 4S_1$ - n

$4S_3$ = n^4 + 6*n(n+1)(2n+1)/6 - 4*n(n+1)/2 + n

$4S_3$ = n^4 + n(n+1)(2n+1) - 2n(n+1) + n

$4S_3$ = n^4 + n(n+1)[(2n+1) - 2] + n = n^4 + n(n+1)(2n-1) + n

$4S_3$ = n[n^3 + (n+1)(2n-1) + 1] = n[n^3 + 2n^2+n-1+1] = n[n^3+2n^2+n]

$4S_3$ = n^2[n^2 + 2n + 1] = n^2(n+1)^2

Result: S_3 = n^2(n+1)^2/4 = [n(n+1)/2]^2 = (S_1)^2

REMARKABLE: Sum of cubes = square of sum of natural numbers!

Verification: 1^3+2^3+3^3 = 1+8+27 = 36 = (1+2+3)^2 = 6^2 [check!]

SUMMARY OF THREE STANDARD RESULTS:

S_1 = Sum(r=1 to n) r = n(n+1)/2

S_2 = Sum(r=1 to n) r^2 = n(n+1)(2n+1)/6

S_3 = Sum(r=1 to n) r^3 = [n(n+1)/2]^2

Key relation: S_3 = (S_1)^2 — this appears frequently in JEE!

2. PRODUCT SERIES AND Vn METHOD

2.1 Sum of r(r+1)

Direct method: r(r+1) = r^2 + r

Sum(r=1 to n) r(r+1) = S_2 + S_1 = n(n+1)(2n+1)/6 + n(n+1)/2

= n(n+1)/6 * [(2n+1) + 3] = n(n+1)/6 * (2n+4) = n(n+1)*2(n+2)/6

Sum r(r+1) = n(n+1)(n+2)/3

2.2 Sum of r(r+1)(r+2) — Vn (Vandermonde) Method

Core Idea of Vn Method:

Define V_r = r(r+1)(r+2)...(r+k) (product of k+1 consecutive integers starting at r)

Then: V_r - V_{r-1} = (k+1) * r(r+1)...(r+k-1)

So: r(r+1)...(r+k-1) = [V_r - V_{r-1}]/(k+1) — this telescopes!

For k=2: r(r+1)(r+2)

Let V_r = r(r+1)(r+2)(r+3)

V_r - V_{r-1} = (r+3)*r(r+1)(r+2) - (r-1)*r(r+1)(r+2) [NOT correct; use identity directly]

Correct: V_r - V_{r-1} = r(r+1)(r+2)(r+3) - (r-1)r(r+1)(r+2) = r(r+1)(r+2)*[(r+3)-(r-1)] = 4*r(r+1)(r+2)

=> r(r+1)(r+2) = (V_r - V_{r-1})/4

Sum = (1/4)*[V_n - V_0] = (1/4)*[n(n+1)(n+2)(n+3) - 0]

Sum r(r+1)(r+2) = n(n+1)(n+2)(n+3)/4

General Result (Vn Method):

Sum(r=1 to n) r(r+1)(r+2)...(r+k-1) = n(n+1)(n+2)...(n+k) / (k+1)

3. FACTORIAL SUMS USING TELESCOPING

Sum(r=1 to n) r * r!

Trick: write r = (r+1) - 1

r * r! = [(r+1) - 1]*r! = (r+1)*r! - r! = (r+1)! - r!

This is a telescoping difference! Write out all terms:

r=1: 2! - 1!

r=2: 3! - 2!

r=3: 4! - 3!

... ...

r=n: (n+1)! - n!

Sum = (n+1)! - 1! = (n+1)! - 1

Sum(r=1 to n) r(r+1) * (r-1)!

= Sum r(r+1)*(r-1)! = Sum [r+1]*r! - ... [requires careful manipulation]

General principle: always try to express T_r as f(r) - f(r-1) for telescoping.

4. SUMS INVOLVING ODD NUMBERS

4.1 Sum of First n Odd Numbers

Odd numbers: 1, 3, 5, 7, ..., (2n-1)

This is an AP: a=1, d=2, last term = 2n-1

S = n/2 * (first + last) = n/2 * (1 + 2n-1) = n/2 * 2n = n^2

Interpretation: n^2 dots can be arranged as an n x n square.

4.2 Sum of Squares of First n Odd Numbers

1^2 + 3^2 + 5^2 + ... + (2n-1)^2 = Sum(r=1 to n)(2r-1)^2

= Sum(4r^2 - 4r + 1) = 4S_2 - 4S_1 + n

= 4*n(n+1)(2n+1)/6 - 4*n(n+1)/2 + n

= 2n(n+1)(2n+1)/3 - 2n(n+1) + n

= n * [2(n+1)(2n+1)/3 - 2(n+1) + 1]

= n/3 * [2(n+1)(2n+1) - 6(n+1) + 3]

= n/3 * [(n+1)(4n+2-6) + 3] = n/3 * [(n+1)(4n-4) + 3]

= n/3 * [4(n+1)(n-1) + 3] = n/3 * [4(n^2-1)+3] = n(4n^2-1)/3

= n(2n-1)(2n+1)/3

5. SECOND-DIFFERENCE METHOD FOR FINDING T_n

Algorithm:

Given series: a_1, a_2, a_3, a_4, ...

Step 1: Compute first differences D1: a_2-a_1, a_3-a_2, a_4-a_3, ...

Step 2: Compute second differences D2 from D1.

Step 3: If D2 is constant => T_n is a quadratic polynomial An^2+Bn+C

Step 4: Set up 3 equations using T_1, T_2, T_3 and solve for A, B, C.

Example: Series 2, 5, 10, 17, 26, ...

D1: 3, 5, 7, 9 | D2: 2, 2, 2 (constant!)

T_n = An^2+Bn+C; T_1=2, T_2=5, T_3=10

A+B+C=2; 4A+2B+C=5; 9A+3B+C=10

Subtracting: 3A+B=3 and 5A+B=5 => 2A=2 => A=1, B=0, C=1

T_n = n^2 + 1

Sum = S_2 + n = n(n+1)(2n+1)/6 + n

6. WORKED EXAMPLES

Example 1: Find Sum(r=1 to 10) r(r+2)

Solution:

r(r+2) = r^2 + 2r

Sum = Sum(r^2) + 2*Sum(r) [r=1 to 10]

= S_2(10) + 2*S_1(10)

= 10*11*21/6 + 2*10*11/2

= 385 + 110

ANSWER: 495

Example 2: Find Sum(r=1 to n) r(r+1)(r+2) using Vn method

Solution:

Define V_r = r(r+1)(r+2)(r+3)

Then V_r - V_{r-1} = 4*r(r+1)(r+2) [as shown in theory]

So r(r+1)(r+2) = (V_r - V_{r-1}) / 4

Summing r=1 to n (telescoping):

Sum = (1/4)*[V_n - V_0] = (1/4)*[n(n+1)(n+2)(n+3) - 0]

ANSWER: n(n+1)(n+2)(n+3)/4

Check n=3: 1*2*3 + 2*3*4 + 3*4*5 = 6+24+60 = 90 = 3*4*5*6/4 = 90 [correct!]

Example 3: Find Sum(r=1 to n) r * r! (factorial sum)

Solution:

Key trick: r = (r+1) - 1

r*r! = [(r+1)-1]*r! = (r+1)! - r!

This is telescoping. Write:

r=1: 2! - 1!

r=2: 3! - 2!

r=3: 4! - 3!

...

r=n: (n+1)! - n!

All middle terms cancel:

ANSWER: (n+1)! - 1

Example 4: Find nth term and sum of: 1, 5, 12, 22, 35, ...

Solution:

Compute differences:

First differences: 4, 7, 10, 13, ... (AP with $d=3$)

Second differences: 3, 3, 3, ... (constant => quadratic T_n)

Let $T_n = An^2 + Bn + C$

$T_1=1$: $A+B+C=1$

$T_2=5$: $4A+2B+C=5$ => subtract (1): $3A+B=4$

$T_3=12$: $9A+3B+C=12$ => subtract (2): $5A+B=7$

Subtracting: $2A=3$ => $A=3/2$

$B = 4 - 3*(3/2) = 4 - 9/2 = -1/2$

$C = 1 - 3/2 + 1/2 = 0$

$T_n = (3n^2 - n)/2 = n(3n-1)/2$

Sum $S_n = (1/2)*(3*S_2 - S_1) = (1/2)*[3n(n+1)(2n+1)/6 - n(n+1)/2]$

$= n(n+1)/2 * [(2n+1)/2 - 1/2] = n(n+1)/2 * n$

ANSWER: $T_n = n(3n-1)/2$, $S_n = n^2(n+1)/2$

Example 5: Find Sum(r=1 to 15) $(2r-1)^2$

Solution:

$(2r-1)^2 = 4r^2 - 4r + 1$

Sum(r=1 to 15) $= 4*S_2(15) - 4*S_1(15) + 15$

$S_2(15) = 15*16*31/6 = 1240$

$S_1(15) = 15*16/2 = 120$

$= 4*1240 - 4*120 + 15$

$= 4960 - 480 + 15$

ANSWER: 4495

Verification using formula $n(2n-1)(2n+1)/3$ at $n=15$:

$= 15*29*31/3 = 15*899/3 = 4495$ [confirmed]

Example 6: Find Sum(r=1 to n) $r^2 * (r+1)$

Solution:

$r^2*(r+1) = r^3 + r^2$

Sum $= S_3 + S_2$

$= [n(n+1)/2]^2 + n(n+1)(2n+1)/6$

$= n^2(n+1)^2/4 + n(n+1)(2n+1)/6$

Factor out $n(n+1)/12$:

$= n(n+1)/12 * [3n(n+1) + 2(2n+1)]$

$= n(n+1)/12 * [3n^2 + 3n + 4n + 2]$

$= n(n+1)/12 * (3n^2 + 7n + 2)$

= n(n+1)/12 * (3n+1)(n+2)

ANSWER: n(n+1)(n+2)(3n+1)/12

7. PRACTICE QUESTIONS — JEE ADVANCED LEVEL

Q1. [Single Correct] The value of Sum(r=1 to 20) r(r+1)(r+2) is:

(A) 47,310

(B) 53,130

(C) 48,510

(D) 50,220

Q2. [Single Correct] Sum(r=1 to n) r * r! equals:

(A) (n+1)! + 1

(B) (n+1)! - 1

(C) n! - 1

(D) (n+2)! - 2

Q3. [Single Correct] 1^2 + 3^2 + 5^2 + ... + 29^2 equals:

(A) 3625

(B) 4495

(C) 4060

(D) 3655

Q4. [Multiple Correct] Which of the following identities are correct?

(A) Sum(r=1 to n) r(r+1) = n(n+1)(n+2)/3

(B) Sum(r=1 to n) r(r+1)(r+2) = n(n+1)(n+2)(n+3)/4

(C) Sum(r=1 to n) r = n(n+2)/2

(D) Sum(r=1 to n) r^3 = [n(n+1)/2]^2

Q5. [Single Correct] For series 2+5+10+17+26+..., sum to n terms is:

(A) n(n+1)(2n+1)/6 + n

(B) n(n^2+2)/3

(C) n^2(n+1)/2

(D) n(n+1)(n+2)/3

Q6. [Single Correct] The value of Sum(r=1 to 10) r(r+2) is:

(A) 440

(B) 495

(C) 550

(D) 605

Q7. [Single Correct] If T_n is nth term of 3+7+13+21+31+..., then T_10 equals:

(A) 103

(B) 111

(C) 121

(D) 131

Q8. [Multiple Correct] Sum(r=1 to 5) r*r! equals: [Select all correct options]

(A) 719

(B) 6! - 1

(C) 720 - 1

(D) 5 * 5!

Q9. [Single Correct] 1*2 + 2*3 + 3*4 + ... + n(n+1) simplifies to:

(A) n(n+1)/2

(B) n(n+1)(2n+1)/6

(C) n(n+1)(n+2)/3

(D) [n(n+1)/2]^2

Q10. [Single Correct] If S_n = Sum(r=1 to n) r^2*(r+1), then S_4 equals:

(A) 110

(B) 120

(C) 130

(D) 140

8. ANSWER KEY WITH SOLUTIONS

Q#	Answer	Solution / Key Step
Q1	(B) 53,130	n(n+1)(n+2)(n+3)/4 at n=20 = 20*21*22*23/4 = 212520/4 = 53130
Q2	(B) (n+1)!-1	r*r!=(r+1)!-r! telescopes; sum=(n+1)!-1
Q3	(B) 4495	Sum(2r-1)^2, r=1 to 15: n(2n-1)(2n+1)/3 = 15*29*31/3 = 4495
Q4	A,B,D	(C) is wrong: correct formula is n(n+1)/2, not n(n+2)/2
Q5	(A)	T_n=n^2+1; Sum=S_2+n = n(n+1)(2n+1)/6 + n
Q6	(B) 495	Sum(r^2+2r) = 385 + 2*55 = 385+110 = 495
Q7	(B) 111	First diffs: 4,6,8,10 => T_n=n^2+n+1; T_10=100+10+1=111
Q8	A,B,C	(n+1)!-1 at n=5 = 6!-1 = 720-1 = 719. (D) 5*120=600 is wrong
Q9	(C)	Product series: Sum r(r+1)=n(n+1)(n+2)/3
Q10	(B) 120	n(n+1)(n+2)(3n+1)/12 at n=4 = 4*5*6*13/12 = 1560/12 = 130. Direct: 1*2+4*3+9*4+16*5=2+12+36+80=130

9. KEY FORMULAS — QUICK REFERENCE

Formula / Series	Closed Form
S1 = Sum r (r=1 to n)	n(n+1)/2
S2 = Sum r^2 (r=1 to n)	n(n+1)(2n+1)/6
S3 = Sum r^3 (r=1 to n)	[n(n+1)/2]^2 = (S1)^2
Sum r(r+1)	n(n+1)(n+2)/3
Sum r(r+1)(r+2)	n(n+1)(n+2)(n+3)/4
Sum r(r+1)(r+2)...(r+k-1)	n(n+1)...(n+k) / (k+1)
Sum r*r! (r=1 to n)	(n+1)! - 1
Sum of first n odd numbers	n^2
Sum (2r-1)^2 (r=1 to n)	n(2n-1)(2n+1)/3
Sum r^2*(r+1) (r=1 to n)	n(n+1)(n+2)(3n+1)/12

ADVANCED NOTES: Sum to N Terms of Special Series

Connections to Binomial Theorem

The Vn method connects to binomial coefficients: Sum r(r+1)...(r+k-1) = n(n+1)...(n+k)/(k+1).

This is equivalent to (k+1)*C(n+k, k+1) = Sum C(r+k-1, k) [Vandermonde-style identity].

JEE Advanced 2019 asked: Sum(r=0 to n) C(n,r)*r = n*2^(n-1) — uses similar ideas.

Second-difference method works for any polynomial sequence — key JEE tool.

Sigma Notation Tricks

Sum(r=1 to n)[f(r+1)-f(r)] = f(n+1)-f(1) [forward telescoping]

Sum(r=1 to n)[f(r)-f(r-1)] = f(n)-f(0) [backward telescoping]

Split: Sum(r^2+r) = Sum(r^2) + Sum(r); distribute sigma over addition.

Scale: Sum(c*f(r)) = c*Sum(f(r)); pull constants out of sigma.

JEE ADVANCED EXAM STRATEGY

Recognise the Type First: Before computing, classify whether it is AP/GP/AGP/Special series.

Use Standard Results: Memorise S1, S2, S3, and the product series formulas — saves huge time.

Telescoping Check: If T_r has product of consecutive integers, try partial fractions first.

AM-GM Shortcut: For min/max with a constraint, always try AM-GM before calculus.

Verification: Always verify with n=1 or n=2 before writing final answer.

AGP Infinite Sum: If |r|<1, use S_inf = a/(1-r) + dr/(1-r)^2 directly.

Nicomachus: Any time S3 appears, immediately replace with (S1)^2.

Multiple Correct MCQs: All four options may be true — verify each independently.

COMPLETE FORMULA CONSOLIDATION

#	Series / Expression	Closed Form
1	Sum r (r=1 to n)	n(n+1)/2
2	Sum r^2 (r=1 to n)	n(n+1)(2n+1)/6
3	Sum r^3 (r=1 to n)	[n(n+1)/2]^2
4	Sum r(r+1)	n(n+1)(n+2)/3
5	Sum r(r+1)(r+2)	n(n+1)(n+2)(n+3)/4
6	Sum r*r! (r=1 to n)	(n+1)! - 1
7	Sum odd numbers (n terms)	n^2

8	Sum (2r-1)^2 (r=1 to n)	n(2n-1)(2n+1)/3
9	Sum (2r)^2 (r=1 to n)	2n(n+1)(2n+1)/3
10	Sum (2r-1)^3 (r=1 to n)	n^2(2n^2-1)
11	1/(r(r+1)) — telescoping	Sum to n = n/(n+1)
12	r/(r+1)! — telescoping	Sum to n = 1-1/(n+1)!
13	AP nth term	a+(n-1)d
14	AP sum n terms	n[2a+(n-1)d]/2
15	GP nth term	a*r^(n-1)
16	GP sum infinity	a/(1-r) for |r|<1
17	AGP sum infinity	a/(1-r) + dr/(1-r)^2
18	1+2x+3x^2+... (|x|<1)	1/(1-x)^2
19	Sum r/2^r (r=1 to inf)	2
20	AM >= GM >= HM	Always; equality iff all equal